



HFC DESTRUCTION IN ARTICLE 5 COUNTRIES

WORKSHOP SUMMARY AND RECOMMENDATIONS
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TABLE OF CONTENTS

Executive summary 3

Workshop discussions 4

Cascade recommendations for HFC destruction projects and field-level data needs 10

Conclusion and acknowledgements 14

Workshop participants 15

EXECUTIVE SUMMARY

Prior to the March 2026 Yale Clean Cooling Conference, Cascade Climate and the [Carbon Containment Lab](#) co-convened a small expert group for a workshop on *"Ensuring effective guardrails and pilots for catalytic HFC destruction projects in Article 5 countries"*. On the heels of a [major private-sector commitment to super pollutant mitigation](#), this workshop discussed how carbon finance could help build capacity to recover and destroy ozone-depleting substances (ODS) and hydrofluorocarbons (HFCs). In particular, it focused on enabling conditions and guardrails for HFC carbon credits in Article 5 countries (i.e., developing countries under the Montreal Protocol). It was motivated by the idea that well-designed carbon finance could help kickstart refrigerant recovery and destruction capacity in countries where these high global warming potential (GWP) gases are traditionally vented, and accelerate progress toward their Montreal Protocol, Kigali Amendment, and domestic regulatory obligations.

During the workshop, participants discussed the opportunity to test the climate effectiveness of these projects - particularly HFC destruction projects - during this voluntary phase, before codifying elements of project best practices into compliance market and policy frameworks. Specifically, participants discussed what guardrails are needed to ensure projects are high-quality; what data should be collected during the voluntary phase to inform future carbon program design; who should be responsible for collecting and evaluating this data at the field level; how to avoid delaying or reducing the ambition of countries' Montreal Protocol obligations; and tradeoffs between reclamation and destruction in Article 5 contexts.

This document summarizes perspectives that emerged in the lead-up to, during, and after the workshop (Part A) [1]. This document also includes Cascade's recommendations for actions the field can take to maximize the value of initial HFC destruction projects as part of the broader refrigerant transition in Article 5 countries (Part B).

Immediate recommendations for near-term HFC destruction projects in Article 5 countries:

1. Prioritize point of origin and full chain of custody data collection and sharing for all HFC gases.
2. Ensure effective price limits are in place for all collected and destroyed HFC gases.
3. Prioritize engagement in countries where early voluntary projects can address identified implementation gaps in national refrigerant transition strategies while supporting future policy and compliance market development.

Medium-term recommendations as HFC destruction projects scale:

4. Build trust between carbon finance and Montreal Protocol stakeholders through collaboration and constructive engagement.
5. Strengthen the ability to cross-compare project-level data with country-level HFC reporting to the Montreal Protocol.
6. Provide clarity on "when" and "how" to transition towards Article 6 carbon programs, national compliance market programs and domestic policy.
7. Further evaluate the role carbon finance should or should not play in supporting HFC reclamation.

[1] Within this document, "participant" feedback is used to describe the perspectives shared by those who attended the workshop. While most of the feedback included throughout this document comes directly from workshop discussions, some perspectives from discussions in the lead-up to and after the workshop are also included.

Additional information on the workshop, and background on the refrigerant management opportunity and carbon market context, can be found [here](#) and [here](#).

MOTIVATION FOR WORKSHOP ORGANIZERS

Cascade Climate is a philanthropy-backed nonprofit. We propel emerging climate solutions from the margins to the mainstream by orchestrating catalytic interventions across science, markets, and policy to overcome systemic bottlenecks. Our super pollutant mitigation efforts aim to build a bridge from a handful of early, catalytically-funded demonstration projects toward scale-up through policies and market mechanisms that support embedding super pollutant reductions within industry value chains and national and subnational policies. Refrigerant management and transition represents our initial priority area within super pollutant mitigation.

PART A: WORKSHOP DISCUSSIONS

Ensuring high-quality HFC destruction projects

The March 5th workshop focused on how to ensure that initial Article 5 HFC destruction carbon projects are delivering verifiable climate impacts, are not unintentionally creating perverse outcomes, and have the ability to provide the reliable, accessible, and available data needed in Article 5 contexts. **If initial projects can meet these criteria, they could play an important role in accelerating the refrigerant transition in Article 5 countries.** Cascade believes the next few years will be critical for demonstrating that carbon finance can effectively catalyze the refrigerant transition.

The workshop focused on several key risk areas that need to be mitigated for projects to be successful:

- **Avoiding perverse incentives:** Avoiding perverse incentives where carbon markets lead to more production or destruction of virgin HFC gas is critical, especially given the [legacy of the HFC-23 scandal](#) more than a decade ago. This is a more material risk for HFC destruction projects than for ODS destruction projects as HFC gases are still being produced during stage-wise phasedown, unlike ODS, which was phased out.
- **Mitigating additionality risks:** This includes what is needed for the field to prove that projects are creating quantifiable improvements to HFC recovery rates that produce real climate impacts beyond the business-as-usual scenario, and are not undercutting regulation. Proving this will require more robust data on baseline venting rates, recovery rates, destruction rates, and recoverable gas volume.

While significant progress has been made toward developing robust methodologies that address these risks at the project level, buy-in across market, industry, and civil society stakeholders will be critical to evaluate the success and impact at the field level. This is especially true as projects scale and begin to have their intended impact - on a country's HFC volumes and HFC-related emissions, and as payments through carbon finance begin to noticeably change technician and industry behaviors.

Workshop topic #1: Perverse Incentives

The problem: Due to HFC destruction credits, used gas could become significantly more valuable than virgin gas.

This could cause virgin gas production to be artificially inflated, or virgin gas to be fraudulently passed off as used gas for destruction. This problem tends to be most acute for high-GWP gases, where the potential value of carbon credits may exceed the price of virgin gas.

SUMMARY OF PERVERSE INCENTIVES FINDINGS

Participants broadly agreed on the value of the proposed solutions for ensuring project quality - particularly the need for clear methodological limits on foundational issues such as eligible gas sources and developer eligibility criteria. While there was general agreement on the value of price limits and chain of custody checks, participants differed on how practical those checks would be to implement in practice and whether the supporting data needed to enforce them would be available.

Detail on the potential solutions discussed to avoid perverse incentives

Limits on refrigerant gas eligibility conditions

- Projects might set limits on what gases are allowed, where or how gases originate, or when gases are eligible. Specific country-level conditions may also be relevant here - for example, excluding or providing additional transparency requirements for projects in countries with active in-country HFC production, which exacerbates increased virgin production risks.
- Participants generally agreed that some of these project eligibility conditions - for example, how to deal with stockpiles - should be put in place and enforced at the methodology level. Several participants also highlighted that it would be beneficial to have a sense for which gas types and sources should be prioritized above and beyond what methodologies deem eligible.

Limits on project developer eligibility criteria

- Registries might set limits on who can conduct such projects - for example, not allowing project developers to also be commercially involved in HFC gas production.
- Participants generally agreed that such eligibility criteria should be put in place and enforced at the methodology level, and that there may be some role for civil society organizations to ensure these criteria are being followed.

Chain of custody checks

- Chain of custody checks help to guarantee the origin of gases and track gases from the point of collection through their eventual destruction. However, chain of custody information may be challenging to acquire, especially if relying on pre-aggregated gas sourcing or working via third parties. Without such information it is more difficult to know the origin of the gases being collected, especially the 'long-tail' of smaller containers and recovery points, which increases the risk of destruction of virgin gas.
- Participants generally agreed that chain of custody checks could be effective in helping to prevent perverse incentives - or at least making potential fraud more explicit - though they varied in their perspectives on the feasibility of collecting chain of custody information for all gases sourced.
 - Some practitioners who work directly with technicians had seen success in collecting this information back to the point of origin and connecting technician payments to their collection of this information, while others advocated for a volume-based threshold for when this information should be required (e.g., only requiring this information for canisters collected above ~50 or ~100kg). There was discussion around how available this information would be as projects

scaled and began working with additional third-party intermediaries, rather than directly with technicians.

- Participants also drew connections to legacy ODS methodologies, which some felt had less stringent chain of custody requirements than those proposed in new HFC destruction methodologies, and questioned how much those methodologies should diverge.
- Several participants mentioned the potential of digital tracking systems - e.g., mobile apps for technicians where they could upload serial numbers and gas collection information in real time - in making these checks feasible and transparent throughout a project's lifespan. Such systems are already being piloted by both refrigerant project developers and governments, and are best practice in a number of similar industries (e.g., consumer packaged goods).

Price limits

- If the price of used gas is kept below the price of virgin gas, there is theoretically no economic benefit to pass off virgin gas as used. However, there may be practical, enforcement, and equity challenges to implementing price limits: pricing is dynamic; price limits may be difficult to enforce; or this payment structure may financially disadvantage local technicians and implementers.
- Participants debated the feasibility of enforcing price limits, both in terms of the availability of this data and how to implement it into project processes.
 - Some practitioners had seen success in collecting quarterly or annual pricing surveys to develop baseline price levels in the market, while others were not collecting this data at all. Most agreed that real-time pricing information could be challenging to collect in the Article 5 context today - though based on Cascade's Article 5 country engagement, some governments and industry players (e.g., equipment manufacturers) do have access to this information.
 - Participants also discussed how such price checks might vary across different gas acquisition models (e.g., who pays technicians and how that price is set); the time thresholds to pass updated pricing information through to projects; and the lack of visibility practitioners often have into gas composition at the point of collection. It is also of note that the same price of kg refrigerant recovered could lead to significant differences in carbon project profitability based on the GWP level of gases (which are sold on a CO₂e basis).
 - Generally, participants agreed that the price paid for recovered gas volumes should be capped below the market rate of virgin gas if and when that data was available, though some practitioners believed this risk was overstated given the challenges to incentivizing recovery at all.

Workshop topic #2: Additionality and policy considerations

The problem: In most Article 5 countries, refrigerants are vented at an extremely high rate today; as such, the baseline expectation is that gases will be vented, and that HFC destruction activities are considered additional (i.e., would not have occurred otherwise).

Over time, additionality may be harder to prove as countries or specific sectors make progress with gas recovery rates. Increasing baseline recovery rates are beneficial for the country's lifecycle refrigerant management (LRM) transition, but necessitate a "graduation" from carbon finance - a transition that can be challenging in practice. If this transition is poorly designed, carbon finance may delay needed regulation or disincentivize policies that reduce venting. All the above will require access to high-quality, reliable, and accessible data as projects scale up, which can be challenging to come by today.

SUMMARY OF ADDITIONALITY AND POLICY CONSIDERATION FINDINGS

Participants broadly agreed on the importance of high-quality primary data for demonstrating additionality - particularly around regulatory enforcement and baseline venting rates - and felt that available data today is inadequate for this purpose. Participants were also encouraged by the potential for future harmonization of project-level and country-level data, particularly as a means of identifying and closing data gaps in the near term.

On the relationship between carbon markets and regulation, participants broadly felt it was important to avoid entrenching an overreliance on carbon finance in ways that could delay regulation and industrial action. At the same time, there was a strong perspective that future carbon market transition mechanisms should not be so restrictive as to disincentivize near-term action or prevent initial projects from getting off the ground.

Detail on the potential solutions discussed to avoid non-additionality or disincentivizing ambitious policy

Demonstrate – don't assume – that regulations don't exist or aren't enforced

- One basic check to ensure regulatory additionality is to show that regulations don't exist for a particular type of gas, or that any regulations that do exist are not enforced. While laws themselves are not definitive - carbon credits can still be additional if a law is not generally enforced - it is important to have evidence that the most likely counterfactual was venting, rather than recovery [2].
- Practitioners generally agreed this was an important requirement, and should be put in place and enforced at the methodology level. However, definitive proof points for demonstrating non-enforcement of regulation may be challenging to come by today. Project developers or registries wishing to confirm non-enforcement may have to rely on expert consultation or imperfect data.

Obtain reliable data on venting rate, recovery rate, and recoverable stocks

- Projects have to show that refrigerant recovery would be unlikely in the absence of carbon finance, and data on venting rate, recovery rate, and recoverable stocks tends to be highly uncertain today. This data can be improved through technician surveys, servicing surveys and field studies, end of life audits, modeling, and other data sources.
- Participants generally felt that data harmonization was an important role for civil society to play. Data is typically collected, aggregated, and reported at the country level. Project developers are not necessarily well-positioned to collect data outside the scope of their particular project. In addition, collecting this data often requires coordination across a range of LRM ecosystem actors in a given country (different government agencies, reclaimers, destruction facilities, original equipment manufacturers (OEMs), servicing companies, etc.).

Cross-compare project data with country-level reporting

- As recovery and destruction capacity scales, aggregated project data on HFC collection sources, volumes, and destruction has the potential to serve as a meaningful cross-comparison against annual Montreal Protocol reporting from a country's Ozone Secretariat. Actors across the carbon market ecosystem tracking destruction volume, venting rates, and recovery rates could not only ensure that

[2] The Integrity Council for Voluntary Carbon Markets (ICVCM) requires that non-enforcement in developing countries be demonstrated with up to date, reliable information (ICVCM 2024 8.2.a.1).

projects are having their intended climate impacts, but also proactively identify gaps in current Kigali reporting structures.

- There was broad interest in exploring how harmonizing project and government country-level data sources could strengthen both over time. Participants were enthusiastic about this as a long-term goal, though several expressed skepticism about near-term feasibility given the limited scale of current projects, the lack of connectivity between data sources, and the typical timelines and granularity of Montreal Protocol reporting data.

Establish a clear transition

- Participants agreed the field needs a clear, credible story for how voluntary carbon market activity “bridges” to durable policy. This story is important to catalytic buyers who want to ensure that early support is complementing, rather than substituting for, long-term climate goals.
- Participants generally agreed on the importance of avoiding long-term reliance on carbon finance as Kigali progress advances. Yet key questions remain on how to design and implement this transition - what timeline is appropriate, what a well-designed “bridge” mechanism looks like, and who should be responsible for implementing it - while ensuring that the path to market for early HFC destruction projects is not prohibitively restrictive so as to forestall near-term action.
- Carbon finance could be bounded by measurable progress benchmarks or conservative, time-based phase-outs. However, if improperly designed, limits like this could be counterproductive. For example, the recovery rate is itself connected to carbon finance - e.g., the rate with carbon credits may be >10%, but could drop to 5% once credits are phased out. Uncertainty around long term project viability could also undermine access to the multi-year financing that projects require.

Roles and responsibilities for ecosystem actors

- Discussions across participants made clear that there is no natural “owner” of field-level data checks today, as opposed to specific project-level obligations that have clear owners (e.g., registries, verifiers and project developers). There is not yet a clear sense for how project-level data, field-level data, and a country’s Montreal Protocol reporting requirements interact within the LRM ecosystem.
- Participants suggested that civil society organizations could help determine who needs to play what role in data collection and field-level data checks as projects begin to scale, and support coordination across ecosystem actors.

Additional insights and perspectives

In addition to the core discussions above, several additional insights and perspectives emerged throughout the workshop:

The role of voluntary carbon market (VCM) activities in Kigali implementation

- Participants with expertise in Montreal Protocol reporting and Kigali Amendment implementation raised important questions about how VCM activities could relate to national reporting systems and country-level phasedown obligations.
- *In the near-term*, there was an understanding that recovery-to-destruction projects supported by voluntary carbon finance can help establish initial recovery infrastructure, improve recovery practices, and build the technical capacity in countries where refrigerants are widely vented. In many Article 5 countries, these capabilities are also important prerequisites for the development of commercially viable reclamation markets and broader LRM systems. In this way, voluntary carbon finance can complement and potentially accelerate a country’s refrigerant transition.
- *For the long-term*, participants noted that questions remain around how destruction activities supported through the VCM should interact with national reporting to the Montreal Protocol. Over time, these activities may need to be integrated into appropriate compliance, policy, or accounting

frameworks to ensure integrity and alignment with national HFC phasedown objectives. Participants also highlighted that it would be important to maintain clear visibility into refrigerant flows, including separate tracking of production and imports, recovery, reclamation, and destruction activities, to avoid ambiguity in reporting and accounting.

Destruction vs. reclamation's role in the refrigerant transition

- Participants from the LRM ecosystem emphasized the potential importance of reclamation within the broader refrigerant transition. While destruction was generally seen as the stronger near-term opportunity for carbon finance in terms of greater confidence in climate impact, reclamation is expected to play a critical role in countries' long-term refrigerant management strategies. Reclamation can help reduce reliance on imported refrigerants, improve supply security, and support the servicing of existing equipment fleets as cooling demand continues to grow. Next-generation refrigerants and equipment are not yet available at sufficient scale in many Article 5 countries, suggesting a two-step refrigerant transition:
 - Step 1: Integrate lower-GWP refrigerants that can be used within existing equipment where appropriate, while strengthening recovery, reclamation, and reuse infrastructure to support servicing needs and improve refrigerant supply security.
 - Step 2: Gradually transition to next-generation equipment and refrigerants over time.
- Participants generally agreed that recovery-to-destruction projects supported by carbon finance could help stand up the recovery infrastructure to enable larger-scale reclamation markets as well.

Technician training and capacity building

- Participants emphasized the importance of working closely with local technicians and servicing companies to build technician capacity through training around proper collection and recovery of these gases. They also encouraged carbon projects to design technician incentive structures that are fit-for-purpose to the country or region in question.
- Participants mentioned that successful models for such training exist and can be replicated, including those supported via the Montreal Protocol Multilateral Fund.

CONTINUE TO NEXT PAGE FOR FIELD-LEVEL RECOMMENDATIONS

PART B: RECOMMENDATIONS

Cascade recommendations for HFC destruction projects and field-level data needs

The refrigerant management field is reaching an exciting and critical inflection point as countries begin to build out their LRM plans, new methodologies emerge, and refrigerant carbon projects begin to operate and scale across Article 5 countries.

Based on the workshop discussions summarized above, and additional on-the-ground learnings with stakeholders across Article 5 countries, **Cascade Climate is putting forward the following field-level recommendations for the initial wave of catalytic, VCM-funded HFC destruction projects.** These recommendations reflect Cascade Climate's synthesis and interpretation of participant insights, as well as our own - they do not necessarily represent the views of individual participants or their organizations.

INITIAL CASCADE RECOMMENDATIONS

Immediate recommendations for near-term HFC destruction projects in Article 5 countries

1. Prioritize point of origin and full chain of custody data collection and sharing for all HFC gases.
2. Ensure effective price limits are in place for all collected and destroyed HFC gases.
3. Prioritize engagement in countries where early voluntary projects can address identified implementation gaps in national refrigerant transition strategies while supporting future policy and compliance market development.

Medium-term recommendations as HFC destruction projects scale

4. Build trust between carbon finance and Montreal Protocol stakeholders through collaboration and constructive engagement.
5. Strengthen the ability to cross-compare project-level data with country-level HFC reporting to the Montreal Protocol.
6. Provide clarity on "when" and "how" to transition towards Article 6 carbon programs, national compliance market programs and domestic policy.
7. Further evaluate the role carbon finance should or should not play in supporting HFC reclamation.

*CONTINUE TO NEXT PAGE FOR FULL RECOMMENDATIONS IN
DETAIL*

Immediate recommendations for near-term HFC destruction projects in Article 5 countries

1) Prioritize point of origin and full chain of custody data collection and sharing for all HFC gases

Visibility back to the point of collection is a critical check to ensure that virgin gases are not being passed off as used. This supply chain traceability is a necessary precondition for guaranteeing high-quality gas sources: as HFC destruction gains attention and financial opportunities grow, so will the risk of fraud. Chain of custody tracking reduces the potential for grey areas or fraud (e.g., due to lack of visibility into the true origin of third-party sourced gases). This guardrail is more critical for HFC gases than for ODS gases, because HFC gases remain in production and virgin gases are typically cheap and readily available.

As such, we believe that HFC destruction methodologies and project developers should prioritize gas collection models that enable full visibility into the point of origin of gases, and should put in place tracking systems that can specify chain of custody documentation extending back to the original point of collection.

We understand that project developers who rely on pre-aggregated gas sourcing via third parties may have less visibility into the origin of gases, especially the 'long-tail' of smaller containers and recovery points. However, these tracking systems are best practice across a number of industries today, and leading Article 5 governments are already experimenting with versions of such tracking software (e.g, the Vietnamese government's [RefNEXT](#) app pilot program). Civil society can support by coordinating common chain of custody approaches across government, industry, and project developers.

2) Ensure effective price limits are in place for all collected and destroyed HFC gases

Price limits are critical to ensure that the price paid for used gas that is destroyed remains below that of virgin gas.

We understand that there is no standard reference price for virgin gas in most markets, prices may fluctuate widely within a given country, and real-time pricing information can be challenging to find and track consistently in Article 5 countries. At the same time, actors such as government entities responsible for Montreal Protocol reporting and leading OEMs in a given country are likely to have at least partial visibility into this data. **Because of the importance of price checks, the onus is on project developers to ensure that their incentive model with technicians is reflective of the price levels in their market, and to seek out the necessary information to prove their payment structure mitigates this risk.** Similarly, we recommend that HFC destruction methodologies put in place reasonable requirements for pricing limits that enable independent public assessment that price arbitrage is not occurring, and encourage civil society organizations to regularly check on these pricing levels and project reporting.

3) Prioritize engagement in countries where early voluntary projects can address identified implementation gaps in national refrigerant transition strategies while supporting future policy and compliance market development

The Article 5 grouping under Montreal covers a wide range of countries at different phases of their development and refrigerant transitions. Countries that have identified financing, implementation, or capacity gaps in achieving their Kigali objectives may benefit from carefully designed early voluntary projects that complement existing efforts. For example, these countries may have large existing refrigerant banks, identified financing or implementation gaps within their Kigali Amendment plans, active government interest in lifecycle refrigerant management, and/or institutional structures that enable coordination across refrigerant policy, carbon market, and Article 6 authorities.

In these contexts, early voluntary projects may help demonstrate implementation models, generate operational data, and support collaboration between governments, industry, and project developers as countries advance their refrigerant transition goals.

Medium-term recommendations as HFC destruction projects scale

4) Build trust between carbon finance and Montreal Protocol stakeholders through collaboration and constructive engagement

Building trust between the carbon finance and Montreal Protocol communities will require sustained dialogue, transparency, and a commitment to constructive engagement from the outset. **Practically, this may mean a willingness from project developers to share chain of custody and pricing data with relevant government stakeholders, including National Ozone Units, and offering their services as technical partners in countries' broader refrigerant transitions - both within and outside of carbon market activities.**

A key part of building this trust is through effective proofs of concept: as early projects demonstrate measurable climate impact, results can be surfaced proactively within Montreal Protocol forums to build familiarity and confidence in carbon finance as a complement to, rather than a substitute for, countries' regulatory obligations. Successful projects can serve as replicable playbooks other countries can adopt, creating a snowball effect of credibility and engagement.

Critically, the data that projects require to generate credits have significant overlap with data that governments ultimately need for their Montreal Protocol reporting (e.g., recovery rates, destruction rates, destruction facility data). Working collaboratively to align these data flows is one of the most practical ways project developers can demonstrate that their goals and those of the Montreal Protocol community are mutually reinforcing.

5) Strengthen the ability to cross-compare project-level data with country-level HFC reporting to the Montreal Protocol

Building on the point above, in order to generate high-quality carbon credits, projects will require granular data on gas collection, aggregation, and destruction. Country-level Montreal Protocol reporting data is based on aggregated reporting from servicing companies, industry actors, destruction facilities, and similar refrigerant ecosystem actors, and can be fragmented and inconsistently reported. Harmonizing these project-level and government data sets can improve country-level reporting, increase visibility into actual gas recovery rates and destruction volumes, and identify gaps in current data reporting structures. LRM carbon market actors can play a role in building data and reporting capacity in the Article 5 countries where they operate. **As such, we encourage project developers and registries,**

to the extent possible, to publicly report and share project data related to gas sourcing and recovery and destruction volumes. A coordinated effort among project developers, carbon registries, governments, and civil society organizations could unlock valuable primary data and enable the cross-comparison of these data sources to identify and close country-level data gaps.

6) Provide clarity on "when" and "how" to transition towards Article 6 carbon programs, national compliance market programs and domestic policy

Early voluntary projects are a catalyst, not a destination. The objective is not long-term reliance on voluntary carbon finance, but the development of durable lifecycle refrigerant management systems supported by industry practice, public policy, and - where appropriate - compliance market mechanisms.

Initial voluntary projects can play an important role in testing project guardrails, generating operational and emission reduction data, and building confidence in approaches to refrigerant recovery and destruction. Over time, these learnings could help inform future consideration of HFC destruction activities within Article 6 mechanisms, domestic compliance markets, and broader refrigerant policy frameworks.

To support this transition, ecosystem actors should actively engage National Ozone Units, Article 6 authorities, and other relevant stakeholders early in the process. Significant questions remain regarding the appropriate timing, design, and governance of any transition from voluntary carbon finance to compliance-based approaches. As project experience and data become available, stakeholders should continue to evaluate these questions and refine potential transition pathways in partnership with governments, industry, and the broader Montreal Protocol community.

7) Further evaluate the role carbon finance should or should not play in supporting HFC reclamation

ODS and HFC destruction projects may receive the bulk of voluntary climate finance in the coming years due to confidence in climate impact and their role in helping to stand up destruction capacity. While reclamation methodologies exist and are active in the VCM today, their impact can be [less straightforward](#) due to questions around displaced virgin production and deferred, rather than eliminated, emissions.

Still, governments are expected to prioritize reclamation in their refrigerant transition plans due to its role in mitigating potential supply shortages and price shocks as HFCs phase down. We expect governments will continue to look for opportunities to utilize carbon finance to build out reclamation infrastructure and subsidize reclaimed refrigerant gas costs (e.g., Thailand's recent [T-VER reclamation crediting methodology](#)). **While reclamation will be an increasingly important tool, it should only be supported by results-based climate finance (e.g., via credits generated from direct emissions reductions) where there is a high degree of confidence in the climate impact.**

Early Article 6 transactions provide a potential model for how to integrate both credited and non-credited activities within a single methodology: for example, in [Ghana's Green Cooling Programme](#), which is supported by Switzerland's KliK Foundation, some mitigation activities are directly credited, while activities like technical capacity building are supported through complementary means under the same project umbrella. Cascade is exploring whether it might be possible to integrate funding support for reclamation capacity in a similar fashion, and it is critical for LRM ecosystem actors to continue to explore the right mix of financing options for reclamation.

Additional future considerations

Gas sourcing principles

In addition to the above recommendations, Cascade believes there may be value in providing additional guidance for gas sourcing beyond what is mandated within carbon methodologies. Such principles can be referenced by buyers seeking clarity on which types of ODS and HFC gases and sources to prioritize for destruction via carbon markets (e.g., how to evaluate higher-vs-lower GWP gases; how to weigh gases sources from end-of-life, servicing, stockpiles; and similar).

Cascade, with input from expert organizations and based on the discussions from this workshop, plans to develop an initial version of this framework.

CONCLUSION

Taken together, the above workshop discussions and Cascade recommendations are intended to provide guidance for how voluntary carbon finance can credibly and catalytically support countries' efforts to accelerate HFC recovery and destruction as part of their broader refrigerant transition strategies and Kigali Amendment implementation.

We look forward to working with project developers, registries, civil society organizations, and other LRM stakeholders to advance the refrigerant transition while helping these mitigation activities scale to their maximum climate potential.

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